

NETWORKS AND COMPLEXITY

Solution 13-8

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 13.8: Fixed points and the logistic map [Lvl. 3]

A famous map is the so-called logistic map

$$x_{i+1} = px_i(1 - x_i)$$

where p is a parameter. Find the fixed points of this map.

Solution

We need to solve

$$x^* = px^*(1 - x^*) \quad (1)$$

Since every term contains a factor of x^* , one solution is

$$x_1^* = 0. \quad (2)$$

To search for other solutions, we divide the equation by x^* , which yields

$$1 = p(1 - x^*) \quad (3)$$

$$1/p = 1 - x^* \quad (4)$$

$$1/p - 1 = -x^* \quad (5)$$

$$(p - 1)/p = x^*. \quad (6)$$

Hence, the second fixed point is

$$x_2^* = \frac{p - 1}{p} \quad (7)$$

We check these results by substituting them into the map. For the first fixed point, we find

$$f(x_1^*) = p \cdot 0 \cdot (1 - 0) = 0 = x_1^*. \quad (8)$$

So, we have shown that $x_1^* = 0$ remains stationary under the action of the map. It is indeed a fixed point.

Similarly, for the second fixed point, we check

$$f(x_2^*) = p \frac{p-1}{p} \left(1 - \frac{p-1}{p} \right) \quad (9)$$

$$= (p-1) \left(1 - \frac{p-1}{p} \right) \quad (10)$$

$$= (p-1) - \frac{(p-1)^2}{p} \quad (11)$$

$$= \frac{p(p-1) - (p-1)^2}{p} \quad (12)$$

$$= \frac{(p-1)(p - (p-1))}{p} \quad (13)$$

$$= \frac{(p-1)}{p} (p - p + 1) \quad (14)$$

$$= \frac{(p-1)}{p} = x_2^*, \quad (15)$$

which confirms that also x_2^* is indeed a fixed point.